Production of Bio-Fuel from Cucumber Skin

Professional Practice – II Final Report

Submitted by

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of

BACHELOR OF TECHNOLOGY

in

PETROLEUM ENGINEERING

DEPARTMENT OF PETROLEUM ENGINEERING

SCHOOL OF ENGINEERING

Under the Guidance of
Dr. Sourav Mukherjee

GAIN MORE KNOWLEDGE
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PRESIDENCY UNIVERSITY

2023-2024

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CERTIFICATE

This is to certify that this report "Production of Bio-Fuel from Cucumber Skin" is a bonafide work of "SHETTY RACHAN BALAKRISHNA (20201PET0004), MUJTBA AAMIR AHMED MOHAMMED (20201PET0015), SIDHARTH MURALI (20201PET0016), and MUHAMMED SAFAL.S (20191PET0064)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in PETROLEUM ENGINEERING during 2023-2024.

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DECLARATION

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ABSTRACT

Production of Bio-Fuel from Cucumber Skin

The increase in world population has stimulated in industrialization therefore increasing the energy demand. Crude oil has been the major source of energy but it is predicted that a drastic decline in crude oil production will take place by the year 2050. Therefore, alternative and renewable energy are being explored in order to satisfy the demand as well as to be environmentally friendly. The solution of this is bio-fuel production. The main objective of this work is to production or extract bio-fuel from *Cucumis sativus* skin which falls under second generation and are eco-friendly. The expected results of the produced bio-fuel are to show similar properties as of conventional ethanol. As well as the purity of the produced bio-fuel is expected to be above 95%.

Key words: Cucumis sativus skin, Alpha-amylase, Renewable Energy, Bio-Fuel, Distillation

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1. INTRODUCTION

Energy demand is extremely high due to the drastic increase in the industrialization as the populationis high. Crude oil has been the major source of energy satisfying these demands. But it's predicted by the year 2050, the crude oil annual production may be decrees drastically beyond the expectation and demand. Therefore, many research are being performed to find an alternate and renewable energy to satisfy the demand and are eco-friendly to the environment.The solution to this is Bio-Fuel.The oil and gas industry are in the process to achieve net zero. Several research has been performed to achieve net zero like CCUS. The solution to add on to the achievement of net zero is by mixing biofuel with diesel in the marine

Diesel engines and any other petrol- and diesel-powered engine. Thereby due to the mixing of biofuel the CO_2 emission is reduced drastically.

Fossil fuels formed from dead plants by exposure to heat and pressure in the Earth's crust were first introduced by Andreas Libavius in the year 1597. After almost 200 years, Mikhail Lomonosov tried to discover and succeeded. Petroleum subsidiaries appreciate many advantages they can be burnt to give energy, which may be used to drive a generator to produce force and smart transportation. They have become safer and less complex over an extended time. Due to the above advantages, the demand for petroleum from ancient times till today never decreased. Fossil fuel emits carbon dioxide (CO2) when it is burnt which is a major greenhouse gas and the primary source of pollution. This has contributed to global warming. They are a non-renewable resource of energy once used they cannot be replaced or reused. Using fossil fuels makes the environment more acidic and harmful this could provoke strange and negative changes in the environment as well as climate and have more impact on human life causing more diseases. For example, coal miners experience the negative impacts of lung diseaseand various breathing issues. The drillers are exposed to chemicals, and mixtures of silica which is dangerous to human health. Oil-based goods supply 90% of the energy used by financially made nations. According to International Energy Agency (IEA), the energy demand is growing up by 40% between the time of 2007 to 2030. The environmental pollution caused by fossil fuels impacts human life as air pollution and other toxins cause illness and death when inhaled. Global air pollution caused

death in 2018 has been estimated at over 8 million people, nearly 1 in 5 deaths worldwide, and according to the National Aeronautics and Space Administration, the carbon dioxide (CO2) released from burning fossil fuels stays in the atmosphere for between 300 to 1000 years.

The need for oil-based goods is directly associated with normal degradation, and climate change as well in the following 50 years we will run out of resources. Along these lines, to meet the essentials we need substitute energy sources that are speedily open, effectively accessible, and environmentally friendly, that is "Biofuels". Biofuel creation from unlimited sources is widely considered the most viable choice rather than oil-based goods and a reasonable technique for regular use.

Biofuel, is any fuel that is obtained from biomass — that is, plant or algae material. Since such feedstock material can be recharged promptly, biofuel is viewed as a wellspring of environmentally friendly power, dissimilar to non-renewable energy sources like petrol,coal, and flammable gas. Biofuel is normally upheld as a financially savvy and naturally harmless option in contrast to oil and other non-renewable energy sources, especially inside the setting of rising petrol costs and expanded worry over the commitments made by petroleum derivatives to an earth-wide temperature boost **[1]**.

During the time Extraordinary Conflict, extended interest for biofuel in light of the extended use as an elective design for imported fuel. That time Germany was the one of the countries had gone through the serious lack of fuel. It was happened when various developments are happening like use of alcohol that was gotten from potatoes and gas. The resulting country likewise prepared grain alcohol mixed in with oil was English. The contention time was prompts the most mechanical changes, but together as one-time unobtrusive oil that set liberated from focus east and inlet countries that reduced off the strain. However, biofuel added away due to the extended store of worldwide and financial interest. During the hour of 1973 and 1979 a serious fuel crisis again hit the various countries. In the non OPEC nations relationship of the petroleum Conveying countries made a significant cut in exchanges. The returns with lack in fills that made a thought towards various energy crisis and the usage of biofuels. The 20th century was made greater lucidity of brain toward the usage of biofuels; the principal avocation behind this was additions costs for oil and interest

in natural improvement as well as the release of the nursery gasses.

1.1 Generations of Bio-fuel production

The bio-fuel is generally divided into three major categories which are described below:[2][3]

1.1.1 First generation bio-fuel

1.1.2 Second generation biofuel

1.1.3 Third generation biofuel

1.1.1 First Generation Bio-Fuel

Energy got from the immediate ignition of natural matter, like wood or harvest deposits. Generally accessible, simple to access and utilize. The first-generation biofuelsproduced from food materials, like corn, sugar stick, sunflower oil, soybeans, starch, and sucrose. They are additionally some of the times alluded to as "customary biofuels."[3]. The primary cycle used to change over these feedstocks into biofuel is maturation. In this cycle, compounds are utilized to separate the carbs in biomass into sugars. These sugars are then changed over into alcohols like ethanol or butanol, which can be utilized as fuel.

In any case, there are additionally a few disservices. One is that they rival food crops for land, water, and different assets. This might possibly drive-up food costs and lead to craving and ailing health.

Another disservice is that plants like corn and soybeans are normally developed with the assistance of petroleum derivatives, pesticides, and composts. They need a ton of energy and assets to develop, so original biofuels are not quite as harmless to the ecosystem as you would naturally suspect.

1.1.2 Second generation Bio-Fuel

Second-age biofuels are gotten from plants that are not utilized for food. They are obtained from things like agrarian waste, wood chips, and grasses. The fundamental cycle used to change over these feedstocks into biofuel is thermochemical transformation. Required heat, pressure, and enzymes are used as external agent to break down the biomass into lignin, cellulose and hemicellulose, which are the main for extraction of bio-fuel [4].The main

advantage of second-generation biofuels is that they have the potential to be more environmentally friendly than first-generation biofuels as they do not compete with food crops for land, water, and other resources. The primary disservice of second-age biofuels is that the cycles used to change over them into biofuel are not too perceived as those for original biofuels. This implies that they are not as proficient and have lower energy yields[1][3][4].

1.1.3 Third generation Bio-Fuel

The third-generation biofuels are newly invented that are produced from internal energy source of algae [5]. The advantage of algae that they can born both salt and sweet waters like lake pond.Beside that the bonus advantage that they are growing rapidly manner without any further treatment.In spite of the above advantages, it was observed that yield value of produced bio-fuel from algae is not satisfied up to the mark [5][6]. Due to less quantity of biofuel produced, it is required cultivate a huge number of algae which may occupy almost more than 50% natural water reservoir. This may affect a huge impact on living being's daily life[6].

1.1.4 Advantages and Disadvantages of Biofuels

❖ **Advantages of Biofuel**

Biofuel diminishes the carbon impression of transportation and various endeavours, exploiting the planet's carbon cycle. One gallon of biofuel overriding one gallon of petrol subordinate reduces ozone draining substance outpourings.

- **Cost-benefit**: Biofuel is priced the same as gasoline on the market today. However, the overall profitability of its use is much higher. It is a cleaner fuel and therefore emits less when burned. As demand for biofuels increases, they may become cheaper in the future.

- **Environmental impact:** When burned, clean biofuels generally emit less particulate matter, sulfur dioxide and airborne toxins than those derived from fossil fuels. Blends of biofuel generally have lower emissions compared to fossil fuels. Combustion of biodiesel can produce slightly higher amounts of nitrogen oxides than petroleum

diesel. In biofuel creation, water is used in enormous sums for washing plants and seeds and for evaporative cooling. In any case, influences nearby area's water supplies. The availability of water resources can limit the production of biofuel crops in countries that might otherwise have a comparative advantage. In areas with low rainfall, the amount of irrigation water needed can be significant. Producing more biofuel crops will also affect water quality.

- **Renewable Energy**: Most fossil fuels will end their life and one day turn to smoke.
- **Easy Supply:** Gasoline fuel is refined from crude oil, a non-endless resource. Current gas holds continue onward for quite a while, but will run out in the near future. Biofuels are made from different sources like compost, crop stores, opposite incidental effects, green development, and plants grew unequivocally for fuel creation.
- **Efficient Fuel:** Biofuel is made from boundless sources and is fairly less ignitable than fossil diesel fuel. It has much better lobbing up properties. It has lower disastrous petroleum product results diverged from standard diesel fuel. Biofuels can be delivered utilizing various materials. The general advantage of direction is much higher.
- **Vehicle engine durability**: Biofuel is sensible for present day engine designs and performs very well much of the time. It has a higher cetane number and better lubricity. Using biodiesel as fuel increases engine life. No engine changes are required. This grants engines to run longer, needless help and reduce as a rule testing costs. Engines expected to run on biofuel have lower releases than other diesel engines.

❖ **Disadvantages of Biofuel**

- **Monoculture:** Monoculture insinuates the show of making equivalent reaps over various years, as opposed to passing different yields on through a rancher's fields after some time. While this may be fiscally enchanting for ranchers at any rate encouraging a similar yield dependably may deny the dirt of upgrades that are returned to the dirt through crop turn. The issues with growing a solitary gather over colossal heaps of land are enormous number. Regardless, developing essentially a lone yield changes the climate to the degree that the food opens to bugs, and they are allowed to obliterate a whole procure. Besides, we could treat the vermin alluded to

above with pesticides, yet a few of those bugs will unpreventably be impervious to the designed materials we use to kill them, and that can have a solitary field of harvests.

- **Using Fertilizer:** Biofuel is made by crops and these yields need compost to foster better. The downside of using manure is that it is damaging to the environment and can cause water defilement. Fertilizers contain nitrogen and phosphorus. They can be cleared out of the soil into adjoining lakes, streams or lakes.

- **Food Shortage:** To Get biofuels from plants and yields with high sugar content. In any case, by far most of these yields are furthermore used as food crops. Vegetable waste can be used as a characteristic substance, yet such food crops are at this point required. It will take command over farmland for various yields which can make a lot of issues.

- **Water Use**: Water arrangement of biofuel crops requires a great deal of water and can descend on neighboring and common water resources if not regulated cautiously. Huge proportions of water are used to convey corn-based ethanol to satisfy neighborhood need for biofuels, which can overburden close by water resource.

2. LITERATUREREVIEW

For the biofuel extraction we need to produce either Bio-diesel or Bio-ethanol or Biogas. Each of them has a different steps and techniques involved for the bio-diesel extraction.

- Multistage trans-esterification of triglycerides from which glycerine produces and isolated using centrifugation.
- Evaluation of glycerol's quality FTIR analysis

For Biogas extraction anaerobic absorption is used. There are various ways for bio-ethanol production but most commonly or most frequently used three ways are:

- Pre-treatment is the chemical procedure that gives simple sugars by converting the complex polysaccharides. Chemical, biological, and physical pre-treatment are the three categories into which the pre-treatment process can be separated. In this process, the biomass arrangement is split into stages to create fermentable sugars.
- During the fermentation stage, a number of chemical and enzymatic reactions produce ethanol by converting the sugar.
- Distillation this separates the pure ethanol from the combination by heating and evaporating until the mixture condenses at the tip of the apparatus. This process produces or collects ethanol.

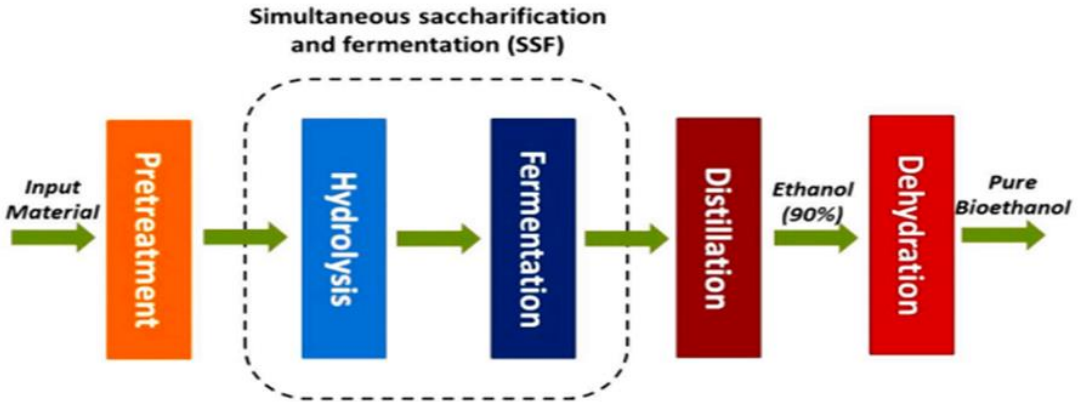


Fig1:Stage wise extraction of Bioethanol from biomass (Achinas. S et.al, 2016)

Several researchers have carried out studies on the first, second and third generation of Bio-fuel. An alternate and renewable energy has to be determined in order to satisfy the energy demands and provide green energy. A brief summary of the studies reviewed are presented in the subsequentsections.

Abdulkareemand his group has focused on the research of production of bio-Fuel from coconut oil[7]. Several experiments and observations were performed to determine the quality and purity of the bio-fuel. The kinematic viscosity at 40°C was 18.17cp and at 100°C was 5.5cp. The total recovery of 98.5% by volume, 0.7% by volume of residue, and 0.8% by volume loss were made. All the properties like flash point, pour point and specific gravity of the sample showed similar characteristics as that of diesel. Osman et al. was carried out research for the production of bio-ethanol from sugarcane using yeast [8]. They observed that optimum conditions for ethanol production was 30°C temperature, with 6 pH.They maintained fermentation medium of 35% sugar cane molasses (20% reducing sugars) supplemented with 1% corn steep liquor as nitrogen source, 1 ml of 24 h yeast culture (0.01 at 600 nm) and shaking rate 200 rpm. Finally, around 13% ethanol was detected through batch fermentation. Mbonu et al. [9]produced bio-fuel from sweet corn using simple distillation apparatus and was properly analyzed. The fundamental aim of this work is to create ethanolfrom starch and sugar as well as contrasting the quality with standard alcohol. The specific objective of this research is to produce ethanol locally from corn as well as to compare the quality with imported ethanol. The Sweet corn was hydrolyzed for 72 hours and later was fermented for 96 hours (4 days). 885g of sweet corn was used. 52.2g of yeast was used in order to increase the rate of fermentation. Result shows that the percentage of ethanol produced is 11.8%. Also, the purity of ethanol produced was determined to be 96%. The highest percentage of purity in aftermarket is 98%.

For the second-generation bio-ethanol production Sharma et.al.[10] has investigated the production of bio-fuel from pretreated banana peel. For every 7 days fermentation was done for banana peels and the ethanol content was measured every 24 hours. The optimum pH and temperature for the fermentation of banana peels was 6 and 30°C respectively. The experimental results indicated that the microwave treatment could lead to significant improvement in sugar production, including reducing sugar and glucose, which provided potential strategies for

widening practical application of banana peel in the food industry. The ethanol produced without pretreated was 5.006% and with pretreated was 6.29%. Michael and his co-researchershas carried out the optimization of production and characterization of bio-fuel produced by enzyme fermentation of hydrolyzed mixed peel (cassava and potato peels)[11]. The optimum conditions for the bio-ethanol yield of 80% by enzymatic (yeast) fermentation are temperature, 46°C, time: 6days, pH :5.7 and enzyme dosage of 2.2ml. The bio-ethanol properties were similar to normal ethanol. Therefore, the result has proved that starch from mixed Peels (Cassava and potato peels) is a good component for bio-ethanol production as biofuel via enzymatic fermentation. Sinem et al. has carried out research about the production of bio-based jet fuel from nettle seed[12]. Due to the increase on oil price over the year have increased fuel price, an alternative fuel for aviation transportation industry from the nettle seed have been purchased from the local market and was obtained using the cold-pressing method at low temperature, after the transesterification process a two phase mixture consisting of biofuel-glycerin was obtained and the upper phase contained fatty acid was taken and transferred to a clean tube, after the final washing process the final bio jet fuel was obtained by adding chemicals at certain rates. They concluded that the produced jet fuel from nettle seed can be a good replacement of today's conventional aviation fuel.

Many researchers[17-24]worked on production of various biofuels types such as Bio-char, Bio-ethanol, Bio-gas, Bio-hydrogen and Bio-methane from cocoa waste and the below table 1 explains in detail.

Table 1: Production of Various Biofuels types from Cocoa Waste

Product	Process	Results
Bio-char	Low-temperature pyrolysis with residence times of 30-120 minutes in 30-minute intervals.	Biochar with a calorific value of 17.8 MJ/kg with high potassium content. The resulting biofuel has characteristics similar to lignite
Bio-char	Pyrolysis at temperatures of 250, 300, and 350°C. Activation process with HCl to reduce the values of free fatty acids to use cooking	The activated carbon generated from CPH showed better retention of free fatty acids than the esterification treatment with H_2SO_4.

	oil used in the production of biodiesel.	
Bio-ethanol	Fermentation using Zymomonasmobilis in a period of 0-8 days. The percentages of microorganisms added were 8-16% v/v.	An alcohol graduation of 10.62% was obtained under 8 day conditions with a concentration of 14% v/v.
Bio-ethanol	Fermentation using Pichia kudriavzevii for 5 days at 30°C	The result shows that the maximum ethanol at 13.8g/L
Bio-ethanol	Saccharomyces cerevisiae was used for fermentation for 12 days.	The bioethanol production was 4.85%; it is reported as a low concentration for the amount of mucilage.
Bio-ethanol	Fermentation with variation in the amount of yeasts during 6 days of the process; hydrolysis pretreatment with H_2SO_4 at temperatures of 30-80°C in periods of 50-150 minutes.	8.46% bioethanol was produced under PH 8 conditions, with a yeast concentration for fermentation of 0.05 mg/g in 6 days of the process. PH was determined to be the main influencing factor.
Bio-Gas	Fermentation process with the application of four pretreatments, acid (H_2SO_4), alkaline (H_2O_2), ground without treatment, and the last one not ground without treatment.	The biogas yields by pretreatment were acid 162.8, alkaline 564.8, ground without treatment 220.8, and unground without treatment 243.3; the highest production was obtained on day 18 of the process.
Bio-Gas	The aerobic digestion process together with a hydrothermal pretreatment	The untreated CPH biogas production was estimated at 3571 (N)/kgVS, while the biogas

	was used at different temperatures (150-220°C) in times of 5-15 minutes.	production from pretreated biomass was 526.38L(N)/kgVS at 150°C for 15 minutes.
Bio-Gas	Fermentation process for 25 days with a pretreatment of a NaOH and NaOH-H_2O_2 solution.	The biogas produced was obtained in a concentration of 66.07% with a yield of 0.734 m3 CH_4/kgVS.
BioHydrogen	Fermentation process at two temperatures 35°C and 55°C; the organic load of volatile compounds 4, 8, and 12 gvs/l was determined.	The amount of gas produced daily was measured. The reaction was faster at 55 °C, the generation of the product was from the fifth day. The load was determined in 12 gvs/l and the hydrogen production of 703 mL
BioHydrogen	Fermentation of the biomass in a load of 10gvs/l at a temperature of 35 °C at a PH of 5.5 in a period of time of 12 days.	The amount of biohydrogen produced was 3674.021 mL, which is equivalent to 91.85 mL H_2/gvs.
Biomethane	Fermentation for the production of biomethane from lignocellulosic material with a pretreatment with N-methylmorpholine-N-oxide at a temperature of 120°C for 3 hours.	The effect of the pretreatment was compared against an untreated control sample, in which the biomethane production increased from 199 to 226 ml of CH_4/gVS. This effect was more relevant for rice straw with an 82% increase.

Demirbas investigated a source of biofuel from algae [13]. Maceiras and his group have also carried out a study to set macro algae as an alternative of vegetable oils for production of bio-fuels.[14]The objective of this paper is to study marine macroalgae as an alternative raw material for the biodiesel production. Macroalgae from the Galician Atlantic coast have been used for biodiesel production using direct transesterification. The goal of this research is to improve the process of producing biodiesel from macroalgae oil by combining extraction with transesterification. The novelty of this research is that, both extraction and reaction of the macroalgae oil are performed simultaneously in the reactor. The process yields 1.6–11.5% depending on the reaction conditions.

Table 2: Properties of microalgae biodiesel [16]

Fuel property	Unit	Microalgae biodiesel	Biodiesel Standard EN 14214	Diesel
CN	–	46.5	51	53.3
Kinematic Viscosity @40 °C	mm^2/s	5.06	3.5–5.0	2.64
Density @15 °C	kg/L	0.912	0.86–0.90	0.84
HHV	MJ/kg	39.86		–
LHV	MJ/kg	37.42		44.0
Acid value	mg KOH/g	0.14	0.5max	0.0
Flash point (closed cup)	°C	95.0	–	71.0
Flash point	°C	–	120	–
Sulphur content	mg/kg	7.5	10max	5.9
Cloud point	°C	16.1	Report	4.0
Lubricity @°60	mm	0.136	–	0.406
Copper corrosion (3 h @50°C)		1a	1 max	1a

Based on the above literatures, it was observed that many researchers performed the production of bio-fuel from different sources. But very few work was done in where researchers used wastage household materials for production of biofuel. After observing those literature, it was concluded that no work was done in where biofuel was produced from cucumber skin. Since the authors were chosen cucumber skin as a raw material for production of biofuel.

Cucumber is a cultivated vegetable throughout the world because of its ability to thrive both

temperature and tropical conditions which result in fresh cucumber throughout the year.

It’s one of the oldest vegetables and believed that it's originated to Asia continent, parts of China with temperate climate and parts of India with tropical climates, at the last cucumber is the fourth most widely cultivated vegetable crop in the world and China is by far the world's largest producer of it.The raw cucumber is 95% water, 4% carbohydrates. 1% protein and contains essential oil, amino acid, pectin, starch, sugars, vitamin C and curcurbita-cin[15]. Due to the presence of starch, sugar and amino acids, the producedbio-ethanol may be a valuable resource for upcoming future.

3. OBJECTIVES

The main objective of this work is to extract bioethanol from cucumber skin. This particular work falls under the category of second-generation Bio-diesel. This work is considered because there is a large number of availabilities of cucumber in the world as well as it is environmentally friendly as we use only the skin of cucumber to produce bio-fuel. The purity of the produced bio-ethanol will be determined using high performance liquid chromatography (HPLC) equipment. The following out comes of the objective are written below in tabulated form

- Extractbio-ethanol from pretreated cucumber skin at different temperature using soxhlet single column distillation at different temperature.
- Determine the presence of alcohol in the extracted liquid by using HPLC.
- Determine the optimum temperature at which presence of alcohol is more.

4. METHODOLOGY:

4.1 MATERIALS:

Raw material for produce bio-ethanol *Cucumis Sativus* was purchased from local market. The hot air oven obtained from Lawrence & Maya India Pvt.Ltd available at the Department of Civil Engineering, Presidency University, Bangaloreis used for remove extra moisture of the collected cucumber skin. Sodium Hydroxide (NaOH)(assay>98%) produced byKarnataka Refinery Pvt. Ltd. was used for neutralization and to remove all the presented fatty acids. Alpha Amylase powder ($C_9H_{14}N_4O_3$) as an enzyme supplied by Karnataka Refinery Pvt. Ltd, is used for accelerate the reaction.The Soxhlet single distillation column was used for distillation for separate ethanol from reaction mixture.

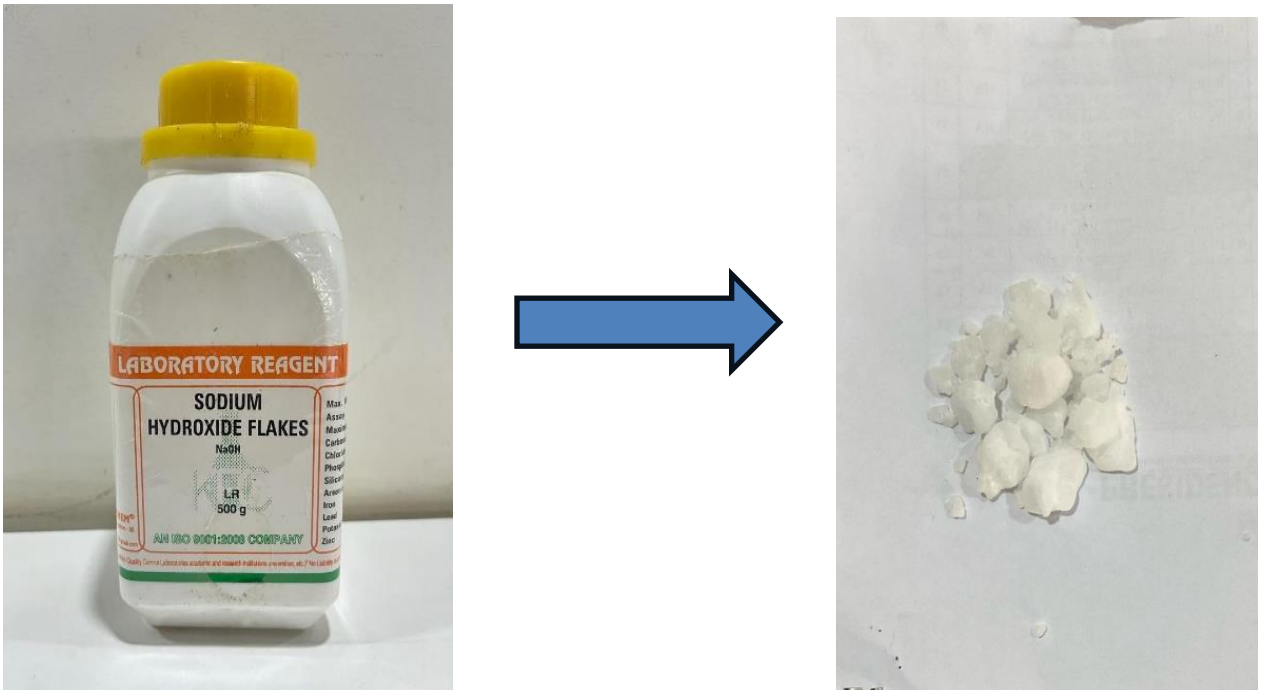


Fig 2: NaOH Flakes

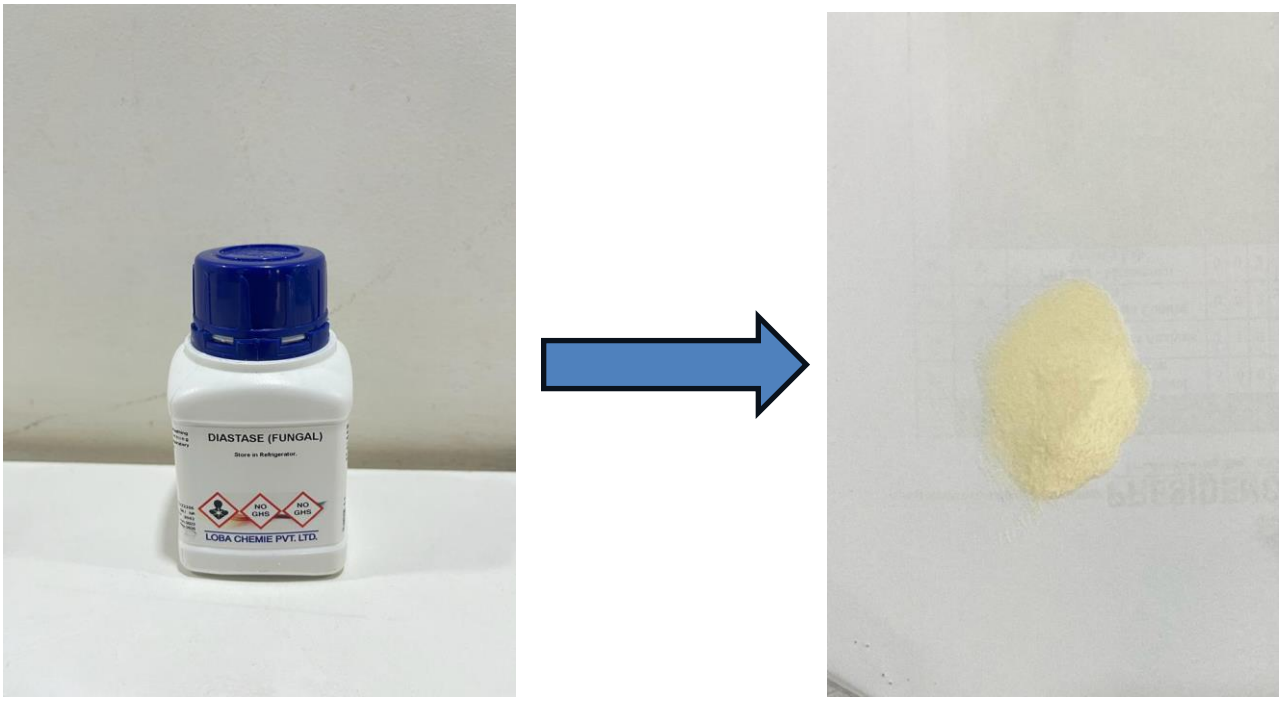


Fig 3: Alpha Amylase Enzyme



Fig4: Soxhlet single distillation column

4.2 EXPERIMENTAL PROCEDURE:

4.2.1 Pretreating of Cucumber skins:

The cucumber skin is collected in a large quantity for about 1-2kgs.The cumber skins were placed in hot air oven at 35℃ for 7days for pretreatment. Pretreated ligno-cellulose is more efficient to produce bio-ethanol than non-pretreated organic compounds. The reason behind the pretreatment is to remove the impurities like unwanted carboxylic acid, minerals and salts. The produced bio-ethanol yield is more than compared to produced bio-ethanol without pretreating. When pretreated, the acid and impurities that is present in the moisture of the cucumber skin is all removed thereby increasing extraction of bio-ethanol.



Fig5: Pretreating of Cucumber skin in Hot Air Oven

Fig 5 shows the Cucumber skin that is placed in the Hot Air Oven. In order to fasten the process of pretreating, the cucumber skin is crushed and placed in the oven in order to dry the sample. The sample is crushed to remove internal moisture, minerals and any salts present in the cucumber skin thereby the efficiency of the extracted Biofuel.



Fig6: Pretreating of crushed Cucumber skin

4.2.2 Neutralization of pretreated Cucumber skins:

The Pretreated cucumber skin was dipped into0.4 (N) NaOH+water solutionto neutralize allorganic acids present in the cucumber skin.This process was done because presence of acid can intervene the fermentation process and it may produce excessive acids than required alcohol. Fig 7 represents the acid removal process from cucumber skin which were dept intoNaOH solution.



Fig7: NaOH mixed solution

4.2.2 Fermentation of pretreated Cucumber skins:

In order to ferment the cucumber skin Alpha Amylase Powder of 3.02g is added into the mixture of cucumber skin and NaOH and left for 7 days. It was observed that after 5-6 days, several bacteria were formed in the mixture as well as heavy foul smell was observed which clearly proved that fermentation was almost done. The Fig 8represents auto fermented cucumber skin after 7 days. In next step this mixture was taken for distillation by using temperature controlledsoxlhet single column distillation unit for separate alcohol from the mixture.



Fig8: Mixture after adding alpha amylase

4.2.3 Single Distillation Column

The extraction of bio-diesel from the fermented mixture of cucumber skin is performed using Single distillation column. The solvent is heated to reflux. The solvent vapor travels up a distillation arm, and floods into the chamber housing the thimble of solid. The condenser ensures that any solvent vapor cools, and drips back down into the chamber housing the solid material. The chamber containing the solid material slowly fills with warm solvent. Some of the desired compound dissolves in the warm solvent. When the Soxhlet chamber is almost full, the chamber is emptied by the siphon. The solvent is returned to the distillation flask. The thimble ensures that the

rapid motion of the solvent does not transport any solid material to the still pot. This cycle may be allowed to repeat many times, over hours or days. Once the vapor is condensed the liquid that is collect using a 500ml beaker is the extracted Bio-Fuel. The extraction was performed with respect to 4 different temperature variations: 75℃, 80℃, 85℃ and 90℃. The temperature range was in between 75℃ to 95℃ because in between this range the bacteria produced alcohol actively.The collected four different extracted bio-fuel submitted for purity testing using HPLC equipment to determine the maximum purity.

4.2.4 High Performance Liquid Chromatography (HPLC)

The HPLC instrument is used to determine the presence of alcohol in the extracted sample. It is a powerful analytical technique used to separate, identify, and quantify components of a mixture. It is widely used in various fields, including pharmaceuticals, chemistry, environmental analysis, and food science.

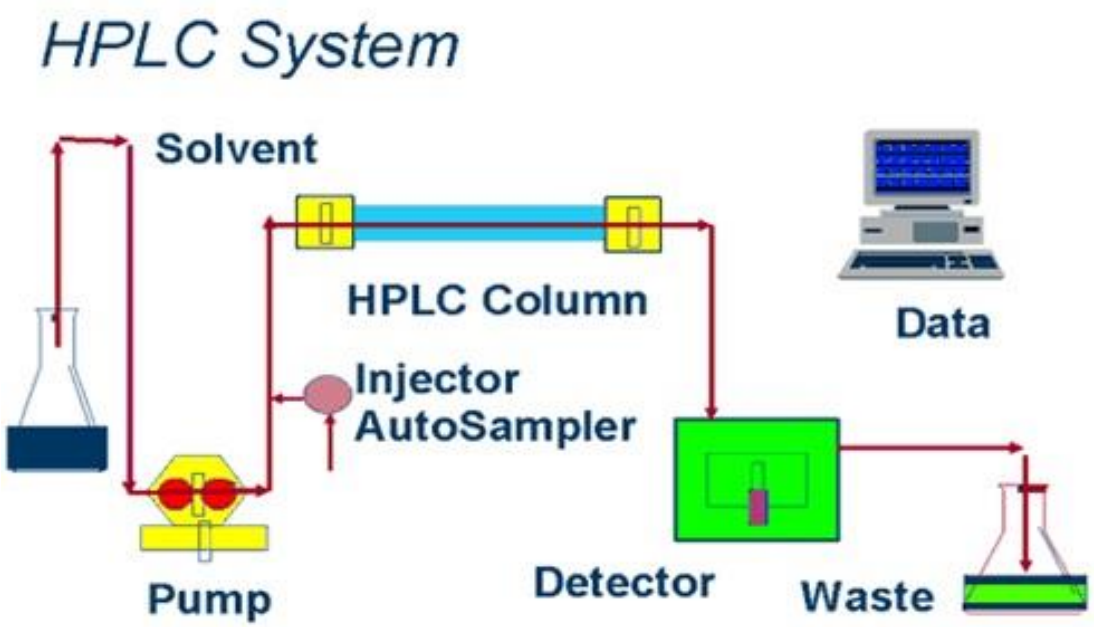


Fig9: Schematic Diagram for HPLC (Upendra, 2013)

- **Sample Preparation:** Prepare the ethanol sample by extracting or diluting the desired components andfiltering out any unwanted particles.

- **Column Selection:** Choose a column that suits the separation needs, considering factors like the nature of the analytes and the desired separation properties.

- **Mobile Phase Preparation:** Prepare a liquid called the mobile phase, which carries the sample through the column. The mobile phase is carefully chosen based on its composition and polarity.

- **Sample Injection and Separation:** Inject the prepared sample into the HPLC system. As the sample moves through the column, different components interact with the column's material, causing them to separate based on their size, polarity, or charge.

- **Detection and Analysis:** As the separated components elute from the column, they pass through a detector that measures their concentration or presence based on specific properties (e.g., UV absorbance). The resulting data is analyzed to quantify the analytes present in the sample.

5. RESULTS AND DISCUSSIONS

5.1 Time taken for 100ml Bio-fuel to collected

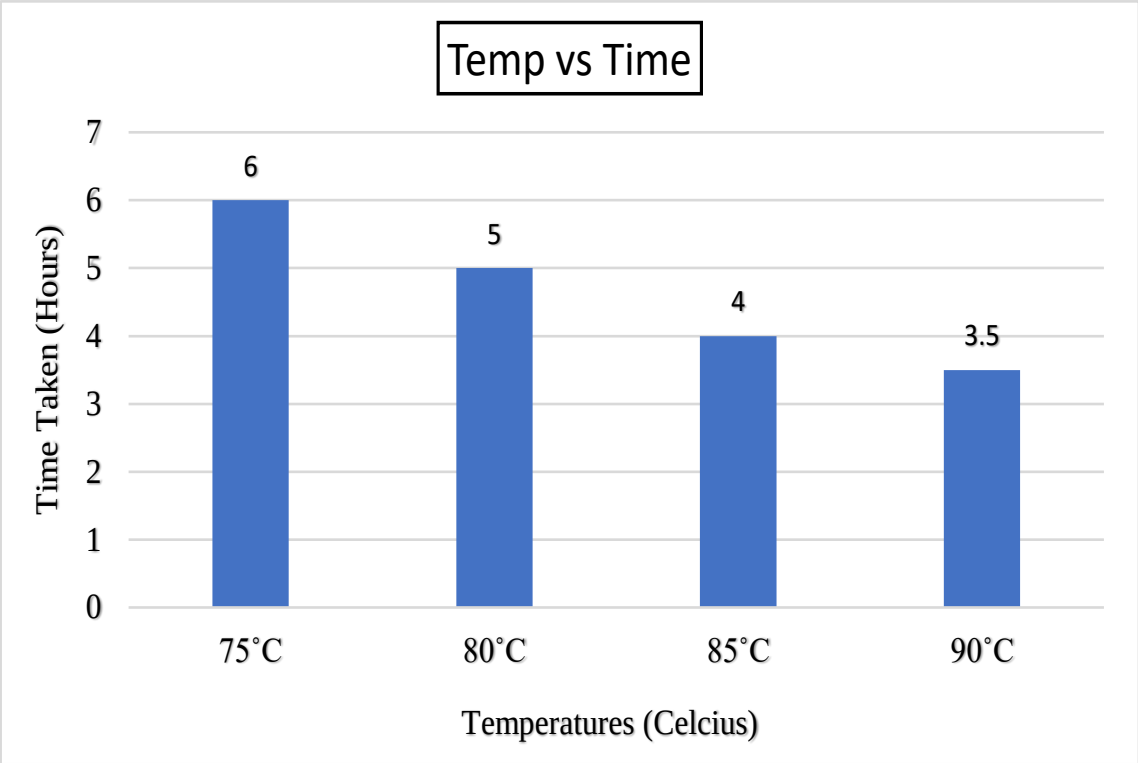


Fig 10: Time vs Temperature graph

- Four different temperature was used to prepare bioethanol from given biomass.
- In each case for collecting bioethanol estimated time was 6 hours, 5 hours, 4 hours and 3.5 hours.
- HPLC process was carried on to check the impurities.

The collected samples are represented in the following where Fig 11 shows the extracted biofuel at 75˚C, Fig 12 shows the extracted biofuel at 80˚C, Fig 13 shows the extracted bio-fuel at 85˚C and Fig 14 shows the extracted Biofuel at 90˚C.



Fig 11: Collected sample at 75˚C



Fig 12: Collected sample at 80˚C



Fig 13: Collected sample at 85 ˚C



Fig 14: Collected sample at 90˚C

5.2 HPLC Data Acquisition

Four different temperature was used to prepare bioethanol from given biomass. The following temperatures are 75˚C, 80˚C, 85˚C, and 90˚C. In each case, for collectingbioethanolestimated time was 6 hours, 5 hours, 4 hours and 3.5 hours. As the temperatures increase time is decreasing and we are getting a high amount of bioethanol at a fast time but for checking the impurities we performed the HPLC test we found the results that showed below.

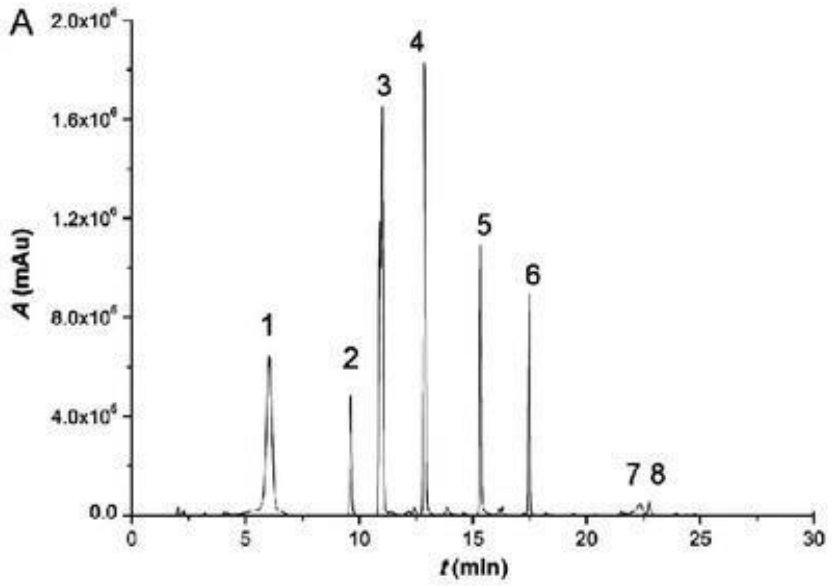


Fig 15: Result of 75˚Csamples after HPLC Test

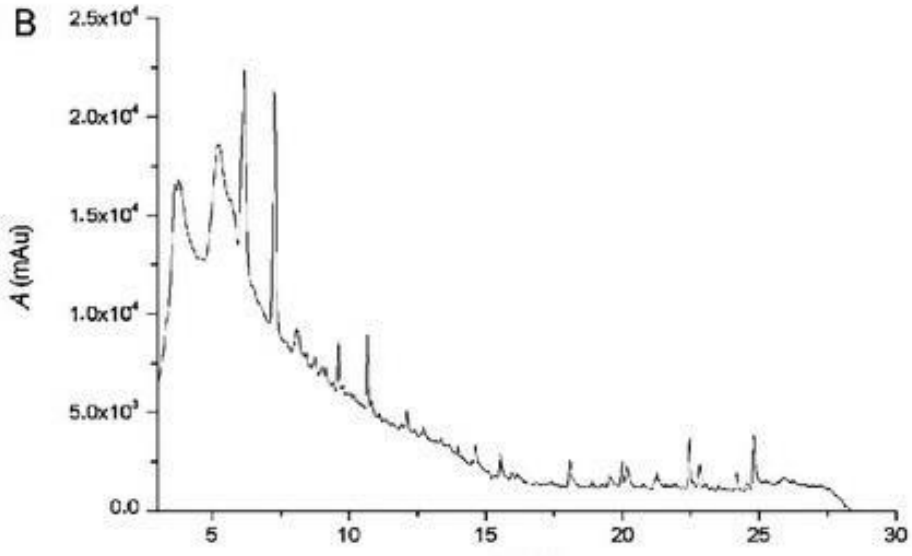


Fig 16: Result of 80˚C samples after HPLC Test

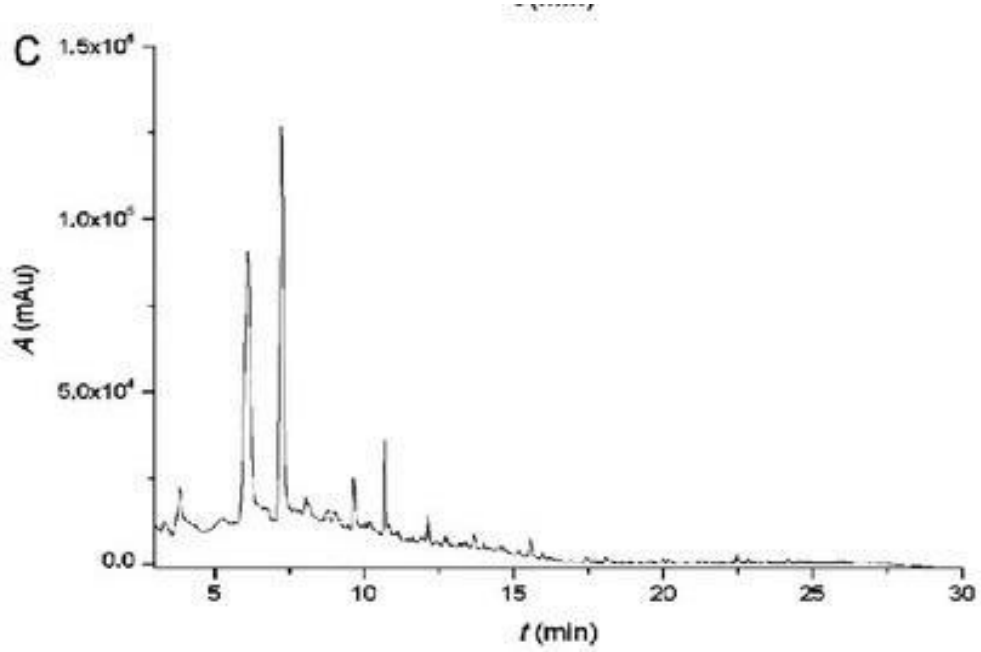


Fig 17: Result of samples 85˚Cafter HPLC Test

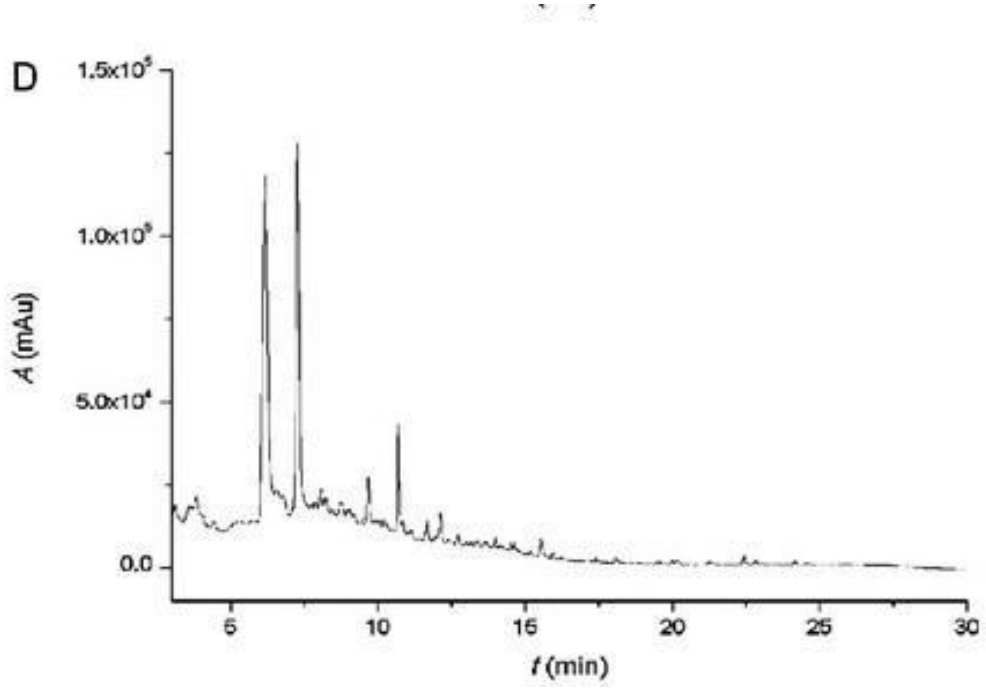


Fig 18: Result of samples 90˚Cafter HPLC Test

The Fig 15, Fig 16, Fig 17 and Fig 18 represents the impurity and the presence of alcohol in the extracted Bio-Ethanol with respect to 75˚C, 80˚C, 85˚C and 90˚C. It is evident that sample A at 75˚C shows the presence of several impurities at the peaks at 6 mins, 9 mins, 11mins, 13mins, 15mins, 17mins, 22 mins and 22.5mins which shows the presence of alcohol is less in the extracted sample A at 75˚C. At 80˚C shows peaks at 2mins, 6 mins, 7mins, 7.5mins, 8.5mins, 11mins, 14mins, 16min, 18mins, 20mins, 22.5mins and 25mins which represents the presence of impurities in the sample B at 80˚C and also shows the presence of alcohol is less. At 85˚C the peaks are very less, 3mins, 6mins, 7.5mins, 9mins, 11mins, 14mins. Therefore, the presence of alcohol is more in this sample C as the peaks at less. Finally, at 90˚Cshows peaks at 2.5mins, 6mins, 7.5mins, 9mins, 11mins, 14mins and 16mins which represents the presence of impurities in the sample D at 90˚C and also shows the presence of alcohol is more comparatively to 75˚C and 80˚C but less than sample C at 85˚C as the straight line is at higher amplitude than that of 85˚C.

6. CONCLUSIONS

6.1 Conclusion

Biofuel is overall kept up with as an economically wise and regularly harmless decision rather than petroleum and other power sources, especially with the current situation of rising petrol costs and broadened worry about the responsibilities made by petroleum power sources to an unnatural weather change. Different express worries about the level of the development of certain biofuels because of the financial expenses related to the refining structure and the expected evacuation of arable land from food creation. It can be concluded that the produced bio-ethanol at 85˚Cis comparatively the better as per HPLC test because the alcohol presence as per the graphs is more compared to other extracted bio-ethanol at 75˚C, 80˚C and 90˚C. Also 90˚C showed better alcohol presence but the amplitude is high when compared to 85˚C.

6.2 SCOPE OF FUTURE WORK

- The purification of extracted bio-ethanol can be performed.
- Efficiency of the extracted and purified bio-ethanol can be purified.
- Comparison of efficiency of produced bio-ethanol with other bio-ethanol produced by other sources can be performed.
- Comparative analysis of bio-ethanol with commercial ethanol could be done.

REFERENCES

[1] Ude, M.U; Oluka, I.S; "Optimization of Production Characterization of Bio-Fuel Produced From Cassava and Potato Peels" *The Proceeding of Nigerian Academy of Science",*Vol.14, 2021

[2] Zheng, C; "Three generation production biotechnology of biomass into Bio-fuel",*GREEN ENERGY AND SUSTAINABLE DEVELOPMENT I: Proceedings of the International Conference on Green Energy and Sustainable Development,* 2017

[3] Bindra, S; Sharma, R; Khan, A; Kulshrestha, S; "Renewable energy sources in different Generation of Bio-fuels with special Emphasis on Microalgae Derived Biodiesel as Sustainable industrial fuel Model" *Biosci.Biotech. Res. Asia,* Vol.14,2017

[4] Ramirez, J; Rainy, T; Brown, A; "A review of hydrothermal liquefaction bio-crude properties and prospects for upgrading to transportation fuel", *Energies,* vol.8, 2015

[5] Li, Y; Wang, B; Wu,N; Lan, C.Q; "Effects of nitrogen sources on cell growth and lipid production of *Neochlorisoleoabundans*" *Appl. Microb. Biotechnol.* Vol.81,2008

[6] Sangita, M; Dilip, K; Brajesh, S; Pravin, K.S; "Biofuels and their sources of production: A review on cleaner sustainable alternative against conventional fuel, in the framework of the food and energy nexus", *Energy Nexus*, 100036, 2021.

[7] Abdulkareem, S; Odigure J. O; Kuranga, M.B; "Production and Characterization of Bio-fuel from Coconut Oil", *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects*, 32:5, 419-425, 2010.

[8] Osman, M.E; Khattab, O.H.; Hammad, I.A;Hussieny, E.I;"Optimization of Bio-Fuel Production by Saccharomyces cerevisiae Isolated from Sugar Cane Bagasse", *Journal of American Science*, 2011,7(5).

[9] Mbonu, O.F; Udeozor, P.A; Umoru, G.U; Uti D;"Production of Bio-fuel from sweet corn (food to fuel)", *Journal of Pharmacognosy and Phytochemistry*, 2016; 5(6): 43-47.

[10] Pallavi, S;Mishra,A.A;"biofuel production from banana peel by using micro wave", *International Journal of Science, Engineering and Technology*, Volume 3 Issue 4:2015.

[11] Michael, U. U; Oluka;"Optimization of production and characterization of bio-fuel produced from cassava and potato", *The Proceedings of the Nigerian Academy of Science*, Volume 14, No 1, 2021.

[12] Sinem; Arzu, O;"Bio- Based Jet Fuel production by transesterification of Nettle seeds",*Engineering, Technology and Applied Science Research*, Volume. 13, No.1, 2023.

[13] Demirbas, A;"Use of algae as biofuel sources", *Energy Conversion and Management* 51 (2010) 2738–2749.

[14] Yameena, M.Z; AlMohamadi;Naqvia, S.R; Noora, T;Chenc, W; Amind, S;"Advances in Production and Activation of marine macroalgae- derived biochar catalyst for sustainable Bio-diesel production", *FUEL* ,Volume 337, 1 April 2023.

[15] Sangita, M; Dilip, K; Brajesh, S; Pravin, K.S; "Biofuels and their sources of production: A review on cleaner sustainable alternative against conventional fuel, in the framework of the food and energy nexus", *Energy Nexus*, 100036, 2021.

[16] Mofiur, M;Rasul, M.G; Hassan. N.M.S; Nabi, M.N "Recent Development In The Production Of Third Generation Biodiesel From Microalgae" *5th InternationalConference On Power And Energy Systems Engineering*, 2018.

[17] Tsai, C.H; Tsai, W.T; Liu, S.C; and Lin, Y.Q; "Thermochemical characterization of biochar from cocoa pod husk prepared at low pyrolysis temperature," *Biomass Conversionand Biorefinery*, vol. 8, no. 2, pp. 237–243, 2018

[18]Billah, M.T;Agratiyan, T. D; Ayu, D;Erliyanti, N. K;Saputro, E. A and Yogaswara, R. R; "Synthesis of bioethanol from cocoa pod husk using zymomonasmobilis," International Journal of Eco-Innovation in Science and Engineering, vol. 1, no. 01, pp.

[19] Roini, C;Asbirayani; Mulya Hartati; and Sundari. A; "Characterization of cocoa pulp (seobroma cacao L.) from south Halmahera as an alternative feedstock for bioethanol production," *IOP Conference Series: Earth and Environmental Science,* vol. 276, no. 1, Article ID 012038, 2019

[20] Awolu, O; and Oyeyemi, S. O; "Optimization of bioethanol production from cocoa (Seobroma cacao) bean shell," *International Journal of Current Microbiology and Applied Science,* vol. 4, pp. 506–514, 2015

[21] Dahunsi, S.O;Adesulu-Dahuns; and Izebere; "Cleaner energy through liquefaction of cocoa ($eobroma cacao) pod husk: pretreatment and process optimization," *Journal of Cleaner Production,* vol. 226, pp. 578–588, 2019

[22] Mancini;Papirio; Lens, P. L. S; and Esposito; "Effect of N-methylmorpholine-N-oxide pretreatment on biogas production from rice straw, cocoa shell, and hazelnut skin*," Environmental Engineering Science,* vol. 33, no. 11, pp. 843–850, 2016

[23] Rojas,J. C;Ram´ırez,J. C; Velasquez,P. E; Acevedo; and Santis, A; "Evaluation of bio-hydrogen production by dark fermentation from cocoa waste mucilage," *Chemical Engineering Transactions,* vol. 79, pp. 283–288, 2020

[24] Widjaja;Nurkhamidah S;Altway, A; Z. Rohmah, A. A; and Saepulah; "Chemical pre-treatments effect for reducing lignin on cocoa pulp waste for biogas production," *AIP Conference Proceedings,* vol. 2349, Article ID 020058, 2021.

[25] Antwi; N; Engler, N; Nelles, M; andSchuch, A; "Anaerobic digestion and the effect of hydrothermal pretreatment on the biogas yield of cocoa pods residues," *Waste Management,* vol. 88, pp. 131–140, 2019.

Photo Gallery

